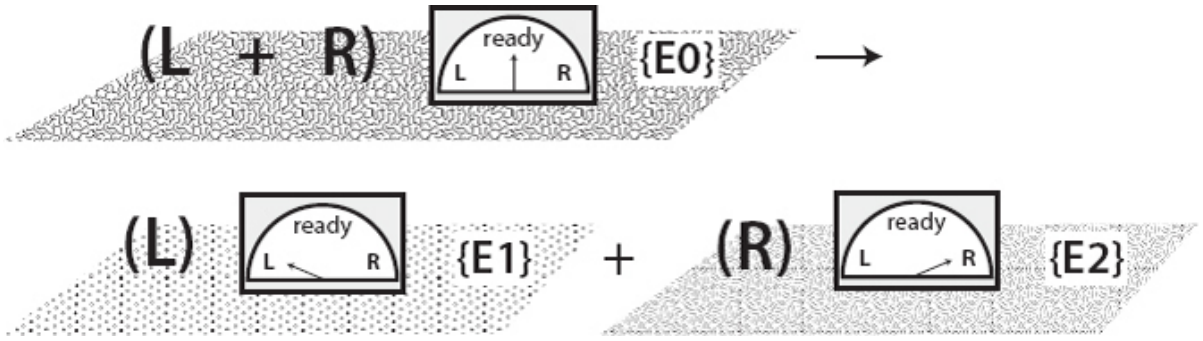


No mysterious collapsing; the whole wave function is still there, evolving cheerfully according to the Schrödinger equation, leaving us in a superposition of two entangled pieces. But note what happens as the electron continues on toward the screen. As before, the state of the electron at any given point on the screen will receive a contribution from what passed through slit L, and another contribution from what passed through slit R. But now those contributions *won't interfere with each other*. In order to get interference, we need to be adding up two equal and opposite quantities:

$$1 + (-1) = 0.$$

But there is no point on the screen where we will find equal and opposite contributions to the electron's wave function from the L and R slits, because passing through those slits entangled the electron with *different states of the rest of the world*. When we say equal and opposite, we mean precisely equal and opposite, not "equal and opposite except for that thing we're entangled with." Being entangled with different states of the detector and environment—being decohered, in other words—means that the two parts of the electron's wave function can no longer interfere with each other. And that means they can't interact at all. And that means they are, for all intents and purposes, part of separate worlds.* From the point of view of things entangled with one branch of the wave function, the other branches might as well be populated by ghosts.

The Many-Worlds formulation of quantum mechanics removes once and for all any mystery about the measurement process and collapse of the wave function. We don't need special rules about making an observation: all that happens is that the wave function keeps chugging along in accordance with the Schrödinger equation. And there's nothing special